

Outlet Intelligence

Unmasking latent demand for 20,000 Sri Lankan beverage outlets

OUTLETS PREDICTED

20,000

HOLD-OUT SPEARMAN ρ

0.793

LKR ALLOCATED

4,999,913

COOLER TOP-100 NET

LKR 31.65 M

The hidden half of demand

HISTORICAL V

Whatever the outlet actually bought.

Constrained, censored, biased downward whenever credit, cooler, or supply got in the way.

LATENT D

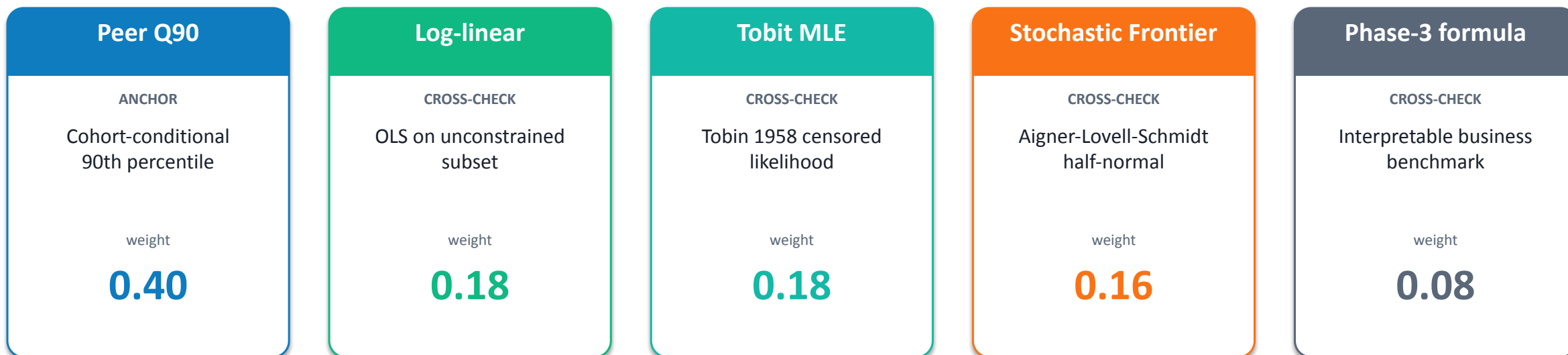
What a healthy version of the outlet would buy in Jan 2026 if constraints were relaxed.

$$V = \min(D, C)$$

observed = min (latent demand , systemic constraint)

"Historical sales tell us what an outlet did sell, not what it could sell. We built the system that closes that gap."

Five textbook methods converge on one answer



Pairwise Spearman ρ — all five rank the outlets the same way

peer vs log-linear 0.86 · peer vs Tobit 0.84 · peer vs SFA 0.85 · peer vs Phase-3 0.84 · log-linear vs Tobit 0.92

No supervised ML. Training on V would reproduce V 's censoring. Five independent textbook methods converge on the same ranking.

43,000 POIs — distance-decay, not flat counts

9 SEMANTIC CATEGORIES

Footfall

σ 150m

School

σ 400m

Tourist

σ 1km

Health

σ 400m

Worship

σ 300m

Food service

σ 150m

Leisure & rec

σ 500m

Population

σ 600m

Competitor

gravity

MARKET STRUCTURE (NON-LINEAR)

HHI · WESTERN PROVINCE (fragmented)

1,101

HHI · NORTH-WESTERN (concentrated)

4,319

TYPE-WEIGHTED COMPETITION PRESSURE

Grocery × Grocery = 1.0

A bus stop 20 m away is not the same as one 400 m away. Gaussian decay for demand drivers, gravity decay for competitors.

How well does it actually generalise?

HOLD-OUT SPEARMAN ρ

(Jan 2025 predicted from 2023-24 only)

0.793

over 10,949 evaluable outlets

SUPPORTING EVIDENCE

Pairwise method agreement

all $\rho \geq 0.83$

5-method ensemble triggered

XAI surrogate fidelity

ρ 0.97 · R^2 0.998

SHAP attributions are faithful

Bit-exact reproducibility

verified

Fresh-clone produces identical CSV

Sanity assertions on output

all passed

Floors/ceilings respected

Hold-out validation pretends Jan 2025 doesn't exist. The rank correlation against actuals is point seven nine.

Where should Western Province spend the budget?

CONCAVE WATER-FILLING LP

maximise $\sum \kappa_i \sqrt{x_i}$ subject to $\sum x_i \leq 5,000,000$ LKR · $0 \leq x_i \leq 5,000$ LKR per outlet

$\kappa_i = \text{uplift} \times \text{cooler_priority} \times (1 + 0.4 \text{ spatial}) / (1 + 0.3 \text{ competition})$

DISCOUNT

LKR 1.74 M

share of LKR 5 M
34.9 %

MERCHANDISING

LKR 2.09 M

share of LKR 5 M
41.8 %

PROMOTIONAL

LKR 1.17 M

share of LKR 5 M
23.3 %

LKR 4,999,912.87 allocated across 8,416 funded outlets.
(172 zero-uplift outlets correctly excluded)

The single strongest recommendation

NET 24-MONTH VALUE

LKR 31.65 M.

derived from a LKR 5 M cooler programme

median payback 3.33 months

CAPEX REQUIRED	24-MONTH MARGIN UPLIFT
LKR 5,000,000 (100 × LKR 50,000)	LKR 36,650,325
NET 24-MONTH VALUE	MEDIAN PAYBACK PERIOD
LKR 31,650,325	3.33 MONTHS
ELIGIBLE OPPORTUNITY	
Outlets currently without a cooler	Total: ~6,500 

Live demo

1 DASHBOARD

- Six KPI tiles
- Province breakdown
- Quick links

2 OUTLETS

- 20,000 outlets
- Six filters
- 20 per page

3 DRILL-DOWN

- Predicted potential
- SHAP drivers
- What-if deltas
- Recommended actions

4 AI EXPLANATION

- Gemini AI
- Grounded in payload
- Plain-English narrative

INSIGHTS WORKSPACE — TABBED VIEWS

Budget · Cooler ROI · Dormancy risk · Distributor scorecard · Territories (Leaflet map) · Forensics findings

From repo to field in 90 days

MONTH 1

PILOT

- Deploy 30 of the Top-100 coolers across two distributors
- Track weekly uplift vs predicted
- Train sales reps on the Outlet Intelligence app
- Deploy silver-clean DQ checks into the live SFA/ERP pipeline

MONTH 2

CALIBRATE

- Fit β (litres-per-LKR elasticity) from pilot data
- Operationalise weekly dormancy alerts
- First distributor scorecard review
- Audit forensic findings in production

MONTH 3

SCALE

- Deploy remaining 70 coolers in the Top-100
- A/B test discount vs merchandising on 200 outlets/channel
- Re-baseline framework with 2026 actuals
- First quarterly trade-spend review driven by this system

"Six months in, every quarterly trade-spend allocation comes from this system."

What this delivers

Capital efficiency

LKR 31.65 M net 24-month value from a LKR 5 M cooler programme

Sales coverage

Top-200 at-risk outlets flagged before they lapse · AUC 0.879

Channel discipline

First per-outlet split of trade spend (34.9 / 41.8 / 23.3)

Operational visibility

10-distributor scorecard · two-standard-deviation best-to-worst gap

Decision integrity

Every recommendation traceable to SHAP driver + counterfactual scenario

The dashboard is live. The Top-100 list is ready. The first cooler can go out next week.